



INTERNATIONAL JOURNAL OF RESEARCH AND STUDIES

Month/Year : April-June, 2026| DOI: <https://doi.org/10.5281/zenodo.22109513>



RECEIVED
27-04-2026



ACCEPTED
12-05-2026



PUBLISHED
30-06-2026

A STUDY TO ASSESS THE PREVALENCE OF WATERBORNE DISEASES AND THEIR IMPACT ON HEALTHCARE SERVICES AMONG THE GENERAL POPULATION IN A SELECTED URBAN AREA OF BILASPUR (C.G.)

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ABSTRACT:

Waterborne diseases constitute one of the most predominant causes of morbidity in developing countries, particularly India, because of inadequately functioning sanitary systems and absence of an adequate water supply. The diseases are caused by bacteria, viruses and parasites and are transmitted through water into the human body through the orofecal route. Typhoid fever, Cholera, Typhoid, Diarrhea, Leptospirosis, Hepatitis A etc. Are some examples. These diseases contribute to a huge personal and health care system burden, in terms of the high volume of hospitalizations, mounting costs of treatments and burden on nursing staff, etc. The present study was conducted on an urban area in Bilaspur district, Chhattisgarh state. This study was descriptive survey, which involved 100 subjects, who were selected by purposive sampling. Data was collected using a structured questionnaire on the socio-demographic characteristics of the subjects, their awareness regarding waterborne diseases and the sources of water, their personal hygiene measures, medical history and their Seekings for medical assistance. The results showed that 74% population were aware about the waterborne diseases very well and 25% people average, 1% poorly. But, about 60% of the population suffered from waterborne disease during the last 1 year. 40% population utilize piped municipal supply, 30% use borewell water, 20% people used purified water and 10% used other sources of water. 50% population visit hospital once or twice a year, while 20% visit hospital five times a year. 65% of the population thought that the cost of treatment was very high, 20% felt that the numbers of cases of the diseases are increasing, while 15% feel that treatment was delayed. Chi-square test revealed that education has a significance association with the awareness with value = 5.6, $P < 0.05$.

Keywords: *Waterborne diseases, healthcare services, prevalence, sanitation, public health, Bilaspur, Chhattisgarh, drinking water, awareness, hospital burden.*

1. Introduction

The element essential for life after air, is undoubtedly water. It requires not only for basic sustenance, but also for cooking, hygiene, etc. Realizing this, in 2010 the United Nations formally designated clean water as a fundamental human right, although a large percentage of the world's population does not have access to clean and safe water.

In the context of waterborne diseases, the water gets contaminated by microbes which then become sources of water borne diseases such as diarrhoea, cholera, typhoid fever, hepatitis A, dysentery, leptospirosis. These are prevalent in developing countries which lacks facilities for water treatment due to lack of adequate infrastructure. As estimated by WHO, about 485,000 deaths can be accounted annually due to unsafe drinking water, with children below the age of five years being more vulnerable to them in low-income countries.

An outstanding example of this situation is India. While the economy has flourished in recent years, the delivery of safe water and sanitation remains elusive to a large portion of the population. Influx of people into cities through urbanization and migration, increase in industrialization, population explosion, agricultural pollution of waterways all contribute to declining quality and availability of water.

In this national picture, Chhattisgarh, and more specifically the city of Bilaspur seems like a very pertinent example. This is one of the most populous cities of the state and experiences widespread waterborne diseases outbreaks due to various reasons including irregular supply of municipal water, over reliance on ground water sources, poor and insufficient sewage disposal systems and garbage accumulation. All these factors combine to create an ideal condition for proliferation and transmission of waterborne diseases.

The overall effects of waterborne disease outbreaks, are not limited to individual victims. Hospitals are flooded with cases which lead to pressure on healthcare systems. The number of beds available and medicines required increases manifold and healthcare professionals face burnout. Waterborne diseases are both a public health and healthcare management problem.

1.1 Background of the Study

The connection between unsanitary water and disease has been known for ages. One major public health event that highlighted this connection was in 1854 when John Snow proved that a cholera epidemic in London was the result of one particular pump being contaminated. This was the first time a direct cause for an epidemic disease has been discovered involving contaminated water. Epidemics of cholera (and other water borne diseases) have wiped out large populations periodically with no concept of how it spread.

Even now, death and disease from waterborne diseases is enormous, primarily in developing countries like India. This is especially so in rapidly urbanising areas for which service provision cannot keep pace. Non-existent piped water supplies, a sewerage system out of order and dependence on shared, open sources (ponds, wells) results in a higher exposure rate.

Bilaspur is the second largest city in Chhattisgarh, in the northern portion of the state, and it contains more than 400,000 people. It is considered a commercial and educational centre of the area; however, the availability of both a potable water supply and adequate sanitation is not up to the level required of an area vulnerable to waterborne diseases.

1.2 Significance of the Study

This study is significant in more ways than one. Firstly, it provides up-to-date, specific data on waterborne diseases and their prevalence in a particular urban area in Bilaspur (C.G.); this data can be used to frame targeted, locally-relevant strategies. Secondly, it looks beyond the disease itself to analyse the effects on health services provision - an area usually ignored by other studies. We need to determine how the disease loads these services so that integrated, sustainable solutions can be developed. Thirdly, the study aims to determine the level of awareness about waterborne diseases. Here it could identify areas where there are knowledge deficits and design relevant health education programs. Fourthly, the relationship between certain socio-demographic characteristics and the disease rate is being analysed; it is here that further variables will be studied, to try and identify potential risk factors in this particular urban community.

1.3 Statement of the Problem

A study to assess the prevalence of waterborne diseases and their impact on health services delivery to the general population in a selected urban area of Bilaspur (C.G.).

1.4 Objectives of the Study

1. To study the socio-demographic characteristics of the respondents.
2. To study the level of awareness of the community regarding waterborne diseases.
3. To study the prevalence of waterborne diseases in the community.
4. To estimate the magnitude of burden of waterborne diseases on health services cost and use.
5. To find out the sources of drinking water and personal hygiene practices of respondents.
6. To determine the association of waterborne diseases with selected socio-demographic variables.

1.5 Hypothesis

H_{0a}: There is no significant relationship between the prevalence rate of waterborne diseases and the chosen socio-demographic factors (age, sex, education, income level and occupation) in the general population in selected urban settlement of Bilaspur.

H_{1a}: There is significant relationship between the prevalence rate of waterborne diseases and the chosen socio-demographic factors in the general population in selected urban settlement of Bilaspur.

H_{0b}: There is no significant relationship between the drinking water sources and prevalence rate of waterborne diseases.

H_{1b}: There is significant relationship between the drinking water sources and prevalence rate of waterborne diseases.

2. Literature Review and Research Gap

The literature review is a review of the body of knowledge available to the topic being researched. The following chapter provides a comprehensive review of various study related to the problem under research concerning national water borne diseases, causes of it, risk factors of water-borne diseases, public knowledge regarding it, and health facility burden of it.

2.1 Global Scenario of Waterborne Diseases

Water-borne diseases is still one of the leading causes of death and suffering globally and largely impacting developing nations. WHO (2020) estimated that some two billion people globally use water contaminated with faces leading to the deaths of thousands every year. Children below the age of 5, including babies, are most vulnerable. According to Reynolds et al. (2004) in a systematic review, more waterborne diseases was found in communities lacking treated water sources. Some waterborne diseases agent that their research identified are E. Coli, Salmonella typhi, Vibrio cholerae, hepatitis A and norovirus. Similarly, press-ups et al. (2019) published a study on global burden of diseases where about 829,000 people die annually from diarrhoeal diseases due to lack of WASH services. A systematic review by Clasen & Haller (2008) also indicated that basic household water treatment such as boiling, filtration and chlorination could significantly reduce diarrhoeal diseases at household level. The UN SDG Progress Report (2021) has also acknowledged that significant progress has been made in achieving SDG 6 (clean water and sanitation) but still billions are deprived from safely managed drinking water services.

2.2 National Scenario of Waterborne Diseases in India

Although India has shown quantifiable improvements in drinking water coverage over the past few decades, the burden of waterborne diseases continues to be a serious problem throughout the country. Park (2005) documented epidemic and endemic outbreaks of cholera, typhoid and diarrhoea across Indian states, where the problem is mostly seen concentrated in areas that lack adequate environmental sanitation facilities, are prone to poverty and lack treated drinking water. Kumar & Singh (2018) have found prevalence of waterborne diseases is highest in slum population in urban India than in the non-slum areas and slum populations have a higher tendency to be admitted to hospitals due to water-borne diseases. Patel (2020) has identified a moderate-to-good level of public awareness of waterborne diseases in urban India, but pointed out the existence of gap of awareness with regards to transmission of diseases and the importance of household water treatment. The study found a positive correlation between levels of education and health awareness, as well as with preventive behaviours. National data are supportive of these observations. Ministry of Health and Family Welfare (2021) has revealed diarrhoea to be among the top causes of hospital admissions in India and millions of such cases reported each year. UNICEF (2020) WASH report acknowledges that while latrine coverage has been massively increased due to Swachh Bharat Mission (SBM) and India is now ODF (Open Defecation Free) in most places but this alone is not sufficient to break the transmission of water-borne diseases.

2.3 State level scenario: Chhattisgarh and Bilaspur

Chhattisgarh has been growing on some parameters, but has been having problems with water and sanitation since its formation in the year 2000. The second biggest city of Chhattisgarh state, Bilaspur functions as a

regional centre for trade, commerce, education and health services but also represents a situation where, due to population growth and aged water distribution systems, waterborne epidemics periodically affect both inhabitants and visitors, visiting for medical treatments in times of epidemic. Rapid urbanization of Chhattisgarh and in particular Bilaspur places additional pressure on water and sanitation systems that were designed to serve far fewer people than they currently do. An aging water distribution system is more prone to leakage than newer systems and this adds risk for contamination. Additionally, Bilaspur is vulnerable during monsoon season that lasts from June to September.

2.4 Causes and Risk Factors of Waterborne Diseases

Smith (2016) has found that lack of access to safe water increases transmission of diseases and contaminated water ingestion poses a significant risk for diseases such as typhoid. Major risk factors for water-borne disease were found to be the lack of treatment of water supply at the source, lack of adequate personal hygiene and poor proximity of the water source to sewage or waste disposal site. Environment plays a significant role. Inadequate drainage infrastructure, stagnant water pools and lack of solid waste management contribute to increase the pathogen load. Density of population coupled with poor sanitation levels increases the incidence rate of infectious diseases.

2.5 Awareness and Prevention of Waterborne Diseases

Awareness and health literacy among communities are extremely important in prevention of waterborne diseases. Kumar & Singh (2018) showed that education level was one of the best predictors of both awareness and health protective behaviour among the study population. Higher levels of education were found to be statistically significant in predicting hand washing practices, as people with secondary education and above were significantly more likely to perform both hand washing and water treatment compared to people with lower levels of education. The effectiveness of health education on attitude toward waterborne diseases was evident through Patel's study (2020) which indicated that health education at community level is associated with change of practices toward waterborne diseases. The guidelines for drinking water quality World Health Organization (WHO) (2020) suggest community involvement and on-going health education as the core elements in the preventive measure.

2.6 Health Facility Burden of Waterborne Diseases

Reynolds et al. (2004) noted a direct link between waterborne disease outbreaks and subsequent hospital admissions. In epidemic conditions, health facilities were observed to face shortages of bed, medicine, and staff, ultimately affecting quality of care. Press-Ups et al. (2019) reported that costs for the treatment of waterborne diseases account for a significant proportion of health expenditure in low- and middle-income countries. Kumar & Singh (2018) further revealed that increasing incidence leads to longer queues of outpatient attendance, prolonged hospital stays and increased cost of individual treatment. The study also points to the psychosocial impact of waterborne diseases on the health workers who are subjected to substantial mental stress and physical fatigue during epidemics.

2.7 Research Gap

A review of the literature on waterborne diseases showed that while the causes and prevention of these diseases has been thoroughly researched, there are several gaps in our understanding, particularly when it comes to its incidence, causes and its burden on health facilities in medium urban centres in Central India. Thus far, previous research has primarily focused on rural or mega urban centres of India with very little focus on cities like Bilaspur. While the increased burden of these diseases on health facilities is recognized in literature, there is a lack of detailed quantitative information on increased patient volumes, treatment costs and service delays at facility level. The majority of available Indian studies used secondary data from older studies that provide outdated findings. The proposed study would help to fill the identified research gaps by providing up-to-date urban data on water-borne diseases in Bilaspur, India. Secondly, while its impact on health care is present in literature, a quantitative study focusing on increase in patient load, cost of treatment and queuing has seldom been conducted. Thirdly, although a large number of studies on water-borne diseases are available in India, but most of them rely on dated resources and very few studies provide the information which is of present use. Therefore, the proposed study aims to fill these gaps

3. Research Design and Methodology

Research Methodology is a term that refers to the scientific process involved in the planning and conducting of research projects. This chapter gives details regarding the methodology use in this research project, which entails the research approach, research design, site, population, sampling, data collection instruments, validity, reliability, and data analysis technique.

3.1 Research Approach

For this study, Quantitative research, in the realm of Quantitative research was applied. Quantitative research is the research used that deals with numerical data. One justification for the use of the quantitative research design in this study is that it focuses on quantifying the prevalence rate of water-borne diseases in the community through numerical values of the number of people that are affected; it quantifies the effect of water borne disease on health delivery services by examining the number of clinic visits due to the disease and expenditure on its treatment; and it also quantifies the relationship between variables under study.

3.2 Research Design

A The research design used was non-experimental descriptive survey design. Descriptive research is a study that is carried out to portray the characteristics of a particular phenomenon or population at a particular time without manipulation of variables by the investigator. This design was chosen because it helps to portray the current situation of water borne disease prevalence in the study community and the subsequent effect it may pose on health services. This study employs survey design in which data was collected from the sample population using a pre-structured questionnaire.

3.3 Data Collection and Variables

The Data that was used to conduct analysis were primary data obtained from a pre-structured self-administrated questionnaire, which was applied to 100 respondents. The choice of the urban residential locality that has high

population density, water problems (supply of clean water) and poor sanitary facilities was due to the problem itself that is already present, high population and access, for the research study to be carried out.

The study population comprised of all adults above the age of 21, residing in the chosen urban locality of Bilaspur municipality. The sample participants were selected using purposive sampling. 100 was chosen as the sample size due to constraints of resources and disease prevalence rates, as it's generally employed in studies of this nature.

Variables used in the model are:

I. Independent Variables: Source of drinking water, treatment of drinking water at house, washing of hands, level of education, average monthly income level of family.

II. Dependent Variables: prevalence rate of water-borne diseases, frequency of hospital visits annually, impact on healthcare delivery system, and treatment costs.

3.4 Description of the Tool

Data was collected through use of a specially designed questionnaire developed by the researcher which was available in both English and Hindi language and comprised of two sections. The first part comprises socio economic parameters such as age, gender, education level, occupation, family income, size of family, religion and marital status while the second section contains information on knowledge of water borne diseases and its modes of transmission, water source, domestic water treatment, hand washing practices, experience of water borne diseases over the last 1 year, use of healthcare services and effect on health care delivery systems.

3.5 Validity, Reliability and Pilot Study

Content validity was established using a validation of questionnaire by five experts in community health nursing, epidemiology and public health. Content Validity Index (CVI) computed was 0.82 indicating good content validity.

Reliability of questionnaire was established by test re-test method and computing Cronbach's alpha coefficient. Correlation coefficient for the two scores set obtained was 0.84 indicating good test-retest reliability. Cronbach's alpha coefficient of knowledge scale was 0.79 indicating good internal consistency. Pilot study was conducted on 10 individuals who are not part of the study sample; this was to see if some questions in questionnaire needs some modifications or changes which are easy to understood.

3.6 Analytical tools

Descriptive. Descriptive statistics (frequencies and percentages) were calculated for all the variables used in the analysis. The Chi-square test for association was employed to identify association between selected socio-demographic parameters and the diseases. A p value of less than 0.05 was taken to be significant. Such statistical tools help us to go beyond raw figures and explore the inter-linkage between selected variables such as level of education, source of water supply and family income.

4. Data Analysis and Findings

The Findings of the study conducted using data obtained from 100 subjects are discussed below. The results are arranged according to the following topics: socio demographic characteristics knowledge and awareness levels; frequency of occurrence of waterborne diseases; source of drinking water and hygiene; effects on health care facilities; and chi-square association test.

4.1 Socio-Demographic Profile of Participants

Understand The socio-demographic profile of the study population becomes critical in understanding the findings and determining those groups that are more vulnerable to risk. The following tables illustrate this diversity, which includes differences in age, education level gender, occupation, and income levels.

Table 4.1: Distribution of Participants According to Socio-Demographic Variables

Socio-Demographic variable	Category	Frequency	Percentage (%)
AGE GROUP	21–30 years	30	30%
	31–40 years	25	25%
	41–50 years	20	20%
	Above 50 years	25	25%
Gender	Male	55	55%
	Female	45	45%
Education	Illiterate	5	5%
	Primary Education	15	15%
	Secondary Education	30	30%
	Higher Secondary (10+2)	25	25%
	Graduate and above	25	25%
Occupation	Government Employee	20	20%
	Private Employee	25	25%
	Self-Employed/Business	20	20%
	Daily Wage Worker	10	15%

Socio-Demographic variable	Category	Frequency	Percentage (%)
	Homemaker	15	15%
Monthly Income	Less than ₹5,000	10	10%
	₹5,001 – ₹10,000	25	25%
	₹10,001 – ₹20,000	35	35%
	₹20,001 – ₹30,000	20	20%
	Above ₹30,000	10	10%
Type of Family	Nuclear Family	60	60%
	Joint Family	35	35%
	Extended Family	5	5%

Source: Primary data collected by the researcher (2025).

The largest number of the participants (30%) was comprised by the people in the age bracket of 21 to 30 years, and that is due to the demographic composition of the chosen urban area. The slightly higher percentage of male participants (55%) could be attributed to the fact that males may be more available during house visits than females. The fairly high educational attainments, in terms of 95% of them having completed their primary education, indicate the urban nature of the study area.

4.2 Level of Knowledge and Awareness About Waterborne Diseases

Table 4.2: Distribution of Participants According to Level of Knowledge About Waterborne Diseases

Level of Knowledge	Frequency	Percentage (%)
Good Knowledge ($\geq 75\%$ score)	74	74%
Average Knowledge (50–74% score)	25	25%
Poor Knowledge ($< 50\%$ score)	1	1%
Total	100	100%

Source: Raw data gathered by the researcher (2025)

The majority (74%) of the respondents surveyed were knowledgeable, and were able to identify the correct information on the causes, the types of disease, methods of transmission, prevention and control. There were 25% average respondent knowledge, that meant they did know, but not entirely. One percent had poor knowledge regarding the subject matter. Although the knowledge level of respondents is promising, the results from the study on disease incidence suggest that good knowledge may not necessarily equate to preventive practices — a phenomenon analogous to what was discussed by Patel (2020).

4.3 Prevalence of Waterborne Diseases

ANALYSING the prevalence of waterborne diseases in the study population reveal a striking disconnect between awareness and actual disease outcomes.

Table 4.3: Comparative Prevalence Analysis — Waterborne Disease Occurrence and Types

Variable	Category	Frequency	Percentage (%)
History of Waterborne Disease (Past 12 Months)	Yes (reported disease)	60	60%
	No (did not report disease)	40	40%
Type of Disease Experienced	Diarrhoea	35	35%
	Typhoid Fever	15	15%
	Gastroenteritis	10	10%
	Dysentery	8	8%
	Hepatitis A	4	4%
	Cholera	3	3%
	More than one disease	5	5%
	Did not report disease	40	40%

Source: Primary data collected by the researcher (2025).

The research stated that 60% of the respondents had one or more waterborne diseases in the past 12 months. High prevalence rate of 60% among respondents is a key observation in the study as these waterborne diseases still represent a major health problem in the selected urban setting despite the knowledge level of the study subjects being high. Most common diseases experienced by respondents were diarrhoea 35%, typhoid fever 15% and gastroenteritis 10%. Low number of respondents with cholera 3% and hepatitis A 4% also still represents health issues.

4.4 Source of Drinking Water and Hygiene Practices

Table 4.4: Sources of drinking water and treatment practices — Water Source and Treatment Practices

Variable	Category	Frequency	Percentage (%)
Source of Drinking Water	Municipal Tap Water	40	40%
	Borewell / Hand Pump	30	30%

Variable	Category	Frequency	Percentage (%)
	Filtered / Purified Water	20	20%
	Packaged Drinking Water	5	5%
	Other Sources	5	5%
Water Treatment at Home	Boiling before drinking	25	25%
	Using water filter/purifier	30	30%
	Adding chlorine/bleach	5	5%
	No treatment (direct use)	40	40%
Handwashing Practices	Always (before eating & after toilet)	40	40%
	Often (most times)	35	35%
	Sometimes (occasionally)	20	20%
	Rarely or never	5	5%

Source: Data generated from the primary research carried out by the researcher (2025).

With regards to sources of drinking water 40% of the respondents sourced drinking water from municipal taps, while another 30% from the bore-wells/hand pumps. The key finding during the research is that 40% of respondents took no action in making their drinking water clean-the proportion is actually alarming as there are chances of contacting water borne diseases as they drink contaminated water. The finding is in corroboration with the 'knowledge action gap' established by Clasen & Haller (2008). The findings that 25 percent of the participants practiced poor hand-washing habits make the scenario even worse.

4.5 The Impact of Waterborne Diseases on Healthcare Services

Similar It is clear from the study results that the waterborne diseases cause a major effect on the healthcare sector in the urban setting of Bilaspur. Due to a high level of diseases prevalence among the selected population, there is an increase in the use of health services, which means an overload of the existing healthcare system.

The study results reveal that a significant share of ill persons have the need for frequent hospitalization. For example, 50% of respondents had been treated once or twice a year, whereas 20% of people had more than five healthcare sessions per year. These results imply that the diseases not only occur but may appear repeatedly because of unsafe water supply and inadequate sanitary conditions.

A government hospital was identified as the main place where a large part of respondents had sought treatment from. It means that governmental hospitals face increased workloads due to the growing number of patients with waterborne diseases. In addition to the effect on the healthcare system, there is also an effect on the household level financially. Most participants said they spent between ₹501 and ₹2,500 during each episode

of treatment, which could be a huge amount for some households with low income. The economic impacts of waterborne diseases in India were illustrated above.

Generally, from the results, it can be seen that waterborne diseases not only harm individuals but also cause many other burdens on the healthcare sector.

Table 4.5: Healthcare Utilization and Impact Metrics

Metric	Category	Frequency	Percentage (%)
Number of Hospital Visits/Year (due to waterborne disease)	1–2 visits	50	50%
	3–5 visits	30	30%
	More than 5 visits	20	20%
Type of Healthcare Facility Used	Government Hospital	45	45%
	Private Hospital/Clinic	35	35%
	Primary Health Centre (PHC)	10	10%
	Pharmacy/Self-medication	8	8%
	Other	2	2%
Perceived Impact on Healthcare Services	Increased patient load in hospitals	65	65%
	Increased treatment cost	20	20%
	Delay in receiving treatment	15	15%
Average Treatment Expenditure per Episode	Less than ₹500	20	20%
	₹501 – ₹1,000	35	35%
	₹1,001 – ₹2,500	25	25%
	₹2,501 – ₹5,000	15	15%
	More than ₹5000	5	5%

Source: Primary data collected by the researcher (2025).

From the data, there is an evident pattern showing a burden of healthcare needs. For patients who have been diagnosed with water-borne illnesses, 50% visit hospitals one or two times in a year, 30% three or five times a year, and alarmingly, 20% visit hospitals more than five times a year. This shows that the condition is reoccurring as well as the potential inadequacy in the treatment of the patient since the patient continues to be

exposed to the source of the illnesses. From the data, it is seen that the utilization rate of government hospitals is at 45%, which proves how important government hospitals are when dealing with such burden. Treatment costs, where 60% spend from ₹501 to ₹2,500 on treatment each time, show great cost for individuals earning from ₹10,001 to ₹20,000 in a month.

4.6 Association Analysis (chi-square test)

Table 4.6: Association Between Key Variables (Chi-Square Analysis)

Variables	Chi-Square (χ^2) Value	Degrees of Freedom (df)	p-Value	Result
Education vs. Knowledge Level	5.6	4	0.03	Significant ($p < 0.05$)
Source of Water vs. Disease Prevalence	6.2	4	0.02	Significant ($p < 0.05$)
Income vs. Treatment Cost	4.8	4	0.04	Significant ($p < 0.05$)
Gender vs. Disease Prevalence	1.3	1	0.25	Not Significant
Age Group vs. Disease Prevalence	3.1	3	0.38	Not Significant

Source: Results of the test- Chi Square test

Correlation between chosen variables and prevalence of waterborne diseases was tested by the chi-square test. It was found out that there is statistically significant correlation between education level and knowledge about waterborne diseases ($\chi^2 = 5.6$, $p < 0.05$). It can be concluded that persons who are better educated tend to know more on the topic.

Moreover, there is significant correlation between drinking water source and waterborne diseases prevalence ($\chi^2 = 6.2$, $p < 0.05$), showing that unsafe drinking water increases risks of having a disease.

Significant correlation between monthly income and costs of treatment was also observed ($\chi^2 = 4.8$, $p < 0.05$), thus pointing at the problem of high cost of treatment and its economic burden on population.

It should be noted, however, that no correlation between gender and waterborne disease prevalence was detected, nor correlation between age group and disease prevalence ($p > 0.05$).

4.7 Summary of Findings

From Our Analysis, we can draw five key conclusions: firstly, despite the fact that the 74% had knowledge on waterborne disease they had experienced waterborne diseases in past 1 year by 60% meaning a vast discrepancy between knowledge and practice; secondly, the most affected disease by our study respondents was diarrhea

(35%), the statistics is in line with the statistics on diarrhea in the country; thirdly, waterborne diseases had placed a burden on healthcare as a whole; fourthly, about 40% did not practice the safe treatment of water and so behaviour change should be given priority; finally, socio-economic aspects are responsible.

5. Limitations of the Study

Several shortcomings need to be addressed when interpreting the results of this research study. A small number of subjects (n=100) might influence the generalizability of the findings of the study. Additionally, the current research was confined to an urban area of Bilaspur which will limit the extrapolation of our results in different areas. Data was only collected at one point in time as the study used a cross-sectional research design. This means that no changes or longitudinal trends in the situation can be observed or monitored over time. The data collected relies on self-reporting from our participants and is therefore subjected to recall bias or even conscious denial of actual situation. Furthermore, laboratory tests for drinking water samples have not been performed on the samples from the study area. Therefore, we cannot establish whether the collected samples were microbiologically contaminated.

6. Conclusion and Future Scope

6.1 Conclusion

This study was conducted with the main aim to evaluate the burden of waterborne diseases and impact of these diseases on healthcare service of the inhabitants of a selected urban area of Bilaspur, Chhattisgarh. Drawing on the responses of a sample of 100 individuals, the results indicate that waterborne diseases continue to pose an active public health problem that impacts the performance of local health services.

The most prominent finding from this investigation was the disparity between respondents' awareness and their behaviours. Although most subjects knew the diseases the spread by waterborne routes and how to prevent it; a significant portion (60%) of them have had one or more waterborne disease in the last year. The results indicate that an external influence (contaminated water and unsanitary conditions) is preventing subjects from taking effective preventive actions.

The chi-square test indicated that there is a statistically significant relationship between the source of drinking water and illness: and water type was one significant part to be the cause of health effects. It is worrisome that an overwhelming proportion of our respondents did not treat their drinking water at home. The association between income and treatment cost also confirms that waterborne diseases exert a tremendous financial burden on poorer households due to the recurrence of treatment cost.

The findings of the study clearly showed that waterborne diseases are the source of stress on healthcare; a substantial fraction of respondents stated that the occurrence of more and more patients has resulted in increased visit at health care and it mainly involves government healthcare institutions. Thus, there is an urgent need to strengthen the public health capacity. The cost of even minimal treatment process becomes a real burden for households particularly the low-income groups.

Finally, it was confirmed that integrated interventions including but not limited to water quality improvement, infrastructure development of sanitary facilities, behaviour change communication and targeted help for vulnerable groups must be involved for efficient solution. It's impossible to tackle waterborne diseases solely through a medical approach; an integrated intervention covering different fields is necessary.

6.2 Future Scope

While we have identified some of the risk factors and burden of disease, future research must be carried out to address specific questions in this domain.

1. Emergence of Novel Waterborne Diseases in the Age of Climate Change:

Given that new challenges to waterborne diseases have arisen with the rise of urbanization and change of the climate, future research will have to focus on rainfall pattern and flooding association to waterborne diseases in Bilaspur.

2. Longitudinal and Seasonal Trends:

With the limited amount of information, we have been able to collect from this cross-sectional study design, further investigation needs to be carried out which will include tracking of trends on a long term and throughout the seasons.

3. Laboratory Assessment of the Microbiological Quality of Water:

In a future research, analysis of water samples collected from different sources where it becomes available for our participants (tap water, bore well water, hand pump etc) need to be investigated by laboratory and microbiological tests.

4. Economic Cost Analysis of Waterborne Diseases:

Further studies need to be done on calculating the economic cost to not just households but to the healthcare sector too in order to highlight the economic benefits of preventing these diseases. Such data is also essential to convince policymakers to take up suitable steps and allocate sufficient resources. In brief, the risks posed by waterborne diseases in Bilaspur present immense opportunities for positive change. The path toward this solution is clearly defined by the evidence from this study. The focus now needs to be shifted from problem identification to problem resolution through coordinated action.

This requires strengthening public health infrastructure, particularly in water supply and sanitation systems. Preventive strategies such as regular water quality testing and timely treatment interventions must be prioritized. Awareness campaigns at the community level can empower individuals to adopt safer hygiene practices. Additionally, improving accessibility and efficiency of primary healthcare services can reduce repeated hospital visits. Inter-sectoral coordination between health, urban development, and water management departments is also essential. Investment in sustainable and long-term solutions will yield both economic and social benefits. Ultimately, a comprehensive and preventive approach can significantly lower disease burden and enhance overall quality of life.

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